

Tim Sippel, Ph.D.

Senior Marine and Aquatic Scientist

I am a senior environmental scientist, with special expertise in marine and aquatic ecology. My diverse experience includes domestic US scientific and regulatory support as well as international projects ranging across South America, Africa, and Asia-Pacific regions. I'm an experienced data scientist with strong technical writing skills, including numerous peer-reviewed publications

Contact

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Employment

Stantec, Senior Environmental Scientist - Fish Ecology SME: Jul 2024 - Present

Position: Senior scientific and analytical support for regulatory compliance projects including hydroelectric dam removals, Biological Assessments, ESA Section 7 consultations, habitat restoration and mitigation, onshore and offshore renewable energy, site remediation, water quality, water management and conveyance, and fishery management. Data analytic and statistical support for diverse projects.

Accomplishments: Contribute to procurement of new business and contracts, mentor junior staff, senior authorship of peer-reviewed publication, member of National Academy of Sciences editorial committee and California Central Valley Improvement Act (CVPIA) Technical team

Worley, Senior Environmental Scientist - Fish Ecology SME: Jun 2022 - Jun 2024

Position: Senior scientific and analytical support for regulatory compliance projects including offshore renewable energy, climate change risk assessment, salmon population assessment, assessment of fishing within global marine protected areas

Accomplishments: Contributed to securing new business and contracts, mentored junior staff, presented at scientific conferences for offshore renewable energy and marine ecology

Washington Department of Fish and Wildlife, Fishery Scientist: Aug 2017 - Aug 2022

Position: Monitoring, forecasting and reporting on salmon population abundance for fishery compliance with the Endangered Species Act and US vs Oregon Management Agreement. Close coordination with other state, Federal and tribal fishery managers. WDFW lead representative to US vs Oregon Technical Team.

Accomplishments: Initiated and lead data modernization and analysis projects to improve transparency, accuracy and efficiency. Contributed new thought leadership to technical issues on data management and analysis with direct application to agency obligations and priorities.

Ocean Associates (NOAA Contractor), Research Biologist: Sep 2011 - Jul 2017

Position: Conducted fishery stock assessments as a member of international and domestic scientific working groups. Served as NMFS representative to the Pacific Fishery Management Council's Highly Migratory Species Management Team advising on fishery management policy.

Accomplishments: Made key scientific contributions to both domestic and international fishery management as stock assessment scientist and scientific advisor. Senior authorship of two peer-reviewed scientific publications.

University of Hawaii, Post-doctoral Scientist: Mar 2010 - Aug 2011

Position: Develop models to estimate behavior of fish carrying satellite transmitters, and contribute to development and deployment of databases to manage satellite tagging data.

Accomplishments: Developed tools for analyses of animal movement ecology, including R codebases and Access databases for large electronic tagging datasets. Co-authored a peer-reviewed book chapter on juvenile white sharks.

Blue Water Marine Research (New Zealand), Fishery biologist: Jan 2004 - Jan 2010

Position: Analyze data, write technical reports, organize and support field research

Monterey Bay Aquarium, Research Technician: Nov 1999 - Jan 2002

Position: Organize and support field and laboratory research

Technical Skills

- **R programming:** tidyverse, time-series, machine learning, visualization
- **GIS:** R packages: sf, terra, leaflet; QGIS
- **Quarto:** Parameterized reporting
- **Data management:** SQL, Git/Github, build and manage relational databases,
- **Specialize Software:** Stock Synthesis

Professional Service

Calif. Central Valley Project Improvement Act, Science Integration Team: 2025 - Present

National Academies of Science, Editorial Committee for the US National Assessment by the Nature Record: 2026

Ministry for Primary Industries - Fisheries New Zealand, Member of Working Groups for Highly Migratory Species and Deepwater Species: 2026 - Present

National Academies of Science, Standing Committee on Offshore Wind and Fisheries Member: 2025 - Committee Dissolved

US vs Oregon Policy Committee, Technical Advisory Committee Member: 2017 - 2022

Pacific Fisheries Management Council, Highly Migratory Species Management Team: 2012 - 2017

International Scientific Committee for Tuna and Tuna-like Species, Stock Assessment Analyst: 2011 - 2017

Project Experience

Fish passage (Oregon); Water quality (Ontario, Canada); Fish passage (Washington); Aquatic habitat improvement (Alaska); Biological Assessment (Oregon); Marine and estuarine habitat monitoring (California); Dam removal (Washington); Salmon spawning habitat supplementation (California); Marine fiber optic cable (Pacific Ocean basin); Aquatic habitat loss mitigation (Peru); Water conveyance (US West coast); Offshore wind feasibility (US West Coast); Marine Protected Area's (Global); Salmon population and fishery impact assessment (Oregon); Offshore wind feasibility (US Pacific Islands); Fishery protection plan (New Jersey); Climate change risk assessment (US & International); Offshore wind feasibility (Australia)

Education

Ph.D. - Marine Ecology, University of Auckland, New Zealand

MSc. - Marine Ecology, Massey University, New Zealand

B.S. - Biology, Kansas State University, Manhattan, KS

Publications

- National Academies of Sciences, Engineering, and Medicine (2026). *Offshore Renewable Energy Development on the West Coast: Understanding Effects on Shipping, Fisheries, and Maritime Activities*. National Academies Press.
- National Academies of Sciences, Engineering, and Medicine (2026). *Review of the Draft U.S. National Assessment by The Nature Record*. The National Academies Press.
- Informal Offshore Wind Energy Group (2024). *Oregon Floating Offshore Wind Energy Roadmap with Exit Ramps*. Oregon Consensus.
- Francis, M., Lyon, W., ... **Sippel, T.** (2023). Post-release survival of shortfin mako sharks released from pelagic tuna longlines in the Pacific Ocean. *Aquatic Conservation: Marine and Freshwater Ecosystems* 33: 366–378.
- Kinney, M., Kacev, D., ... **Sippel, T.** (2020). Common thresher shark *Alopias vulpinus* movement: Bayesian inference on a data limited species. *Marine Ecology Progress Series* 639: 155-167.
- Cox, B. & **Sippel, T.** (2020). Pound Nets and PD17 2017 Survival Analysis. *Washington Department of Fish and Wildlife*: 12.
- Nasby-Lucas, N., Dewar, H., ... **Sippel, T.** (2019). Movements of electronically tagged shortfin mako sharks (*Isurus oxyrinchus*) in the eastern North Pacific Ocean. *Animal Biotelemetry* 7: 12.
- Kacev, D., **Sippel, T.**, ... (2017). An Introduction to Modelling Abundance and Life History Parameters in Shark Populations. *Advances in Marine Biology* 78: 45–87.
- Stohs, S. & **Sippel, T.** (2017). Analysis of Increasing the Required VMS Ping Rate for the California Drift Gillnet Fishery. *NOAA-NMFS*: 18.
- Sippel, T.**, Lee, H., ... (2017). Searching for M: Is there more information about natural mortality in stock assessments than we realize? *Fisheries Research* 192: 135–140.
- Kai, M., Carvalho, F., ... **Sippel, T.** (2017). Stock assessment for the north Pacific Blue shark (*Prionace glauca*) using Bayesian State-space Surplus Production Model. *NOAA*: 43.
- Carvalho, F., Kai, M., ... **Sippel, T.** (2017). Stock Assessment of Blue Shark in the North Pacific Ocean Using Stock Synthesis. *NOAA*: 52.
- Kohin, S., **Sippel, T.** & Carvalho, F. (2016). Catch and Size of Blue Sharks Caught in U.S. Fisheries in the North Pacific. *NOAA - Southwest Fisheries Science Center*: 11.
- Carvalho, F. & **Sippel, T.** (2016). Direct Estimates of Gear Selectivity for the North Pacific Blue Shark Using Catch-at-Length Data: Implications for stock assessment. *NOAA - Pacific Islands Fisheries Science Center*: 10.
- Sippel, T.**, Semba, Y., ... (2016). Size and sex structure of blue sharks in the North Pacific Ocean. *NOAA - Southwest Fisheries Science Center*: 11.
- Holdsworth, J., Saul, P., ... **Sippel, T.** (2016). Synthesis of New Zealand Gamefish Tagging Data, 1975 to 2014. *New Zealand Ministry for Primary Industries*.
- Maxwell, S., Hazen, E., ... **Sippel, T.** (2015). Dynamic ocean management: Defining and conceptualizing real-time management of the ocean. *Marine Policy* 58: 42–50.
- Sippel, T.**, Ohshimo, S., ... (2015). Spatial and temporal patterns of shortfin mako shark size and sex in the North Pacific Ocean. *NOAA*: 12.
- Sippel, T.**, Paige Eveson, J., ... (2015). Using movement data from electronic tags in fisheries stock assessment: A review of models, technology and experimental design. *Fisheries Research* 163: 152–160.
- Sippel, T.**, Carvalho, F., ... (2014). Catches of shortfin mako sharks from US commercial and recreational fisheries in the North Pacific Ocean. *NOAA*: 36.
- Sippel, T.**, Nasby-Lucas, N. & Kohin, S. (2014). Description of the Hawaii Longline Observer Program. *NOAA*: 23.
- Sippel, T.**, Oshimio, S., ... (2014). Spatial and temporal patterns in size and sex of shortfin mako sharks from US and Japanese commercial fisheries: a synthesis to guide future research. *NOAA*: 36.

- Takahashi, N., Kanaiwa, M., ... **Sippel, T.** (2014). Stock Assessment and Future Projections of Blue Shark in the North Pacific Ocean by Bayesian Surplus Production Model using Revised Data. *International Scientific Committee for Tuna and Tuna-like Species in the North Pacific Ocean*: 57.
- Evans, K., Abascal, F., ... **Sippel, T.** (2014). The horizontal and vertical dynamics of swordfish in the South Pacific Ocean. *Journal of Experimental Marine Biology and Ecology* 450: 55–67.
- Sippel, T.**, Wraith, J. & Kohin, S. (2013). A summary of blue marlin conventional tag recapture data from NMFS-SWFSC Cooperative Billfish Tagging Program in the Pacific Ocean. *ISC/13/BILLWG-1/08*.
- Sippel, T.** & Kohin, S. (2013). Catches of blue sharks from U.S. West Coast recreational fisheries during 1971-2011. *NOAA*: 4.
- Urbisci, L., Runcie, R., ... **Sippel, T.** (2013). Examining size-sex segregation among blue sharks (*Prionace glauca*) from the Eastern Pacific Ocean using drift gillnet fishery and satellite tagging data.
- Urbisci, L., **Sippel, T.**, ... (2013). Size composition and spatial distribution of shortfin mako shark by size and sex in US West Coast fisheries.
- Weng, K., O'Sullivan, J., ... **Sippel, T.** (2012). Back to the wild: Release of juvenile great white sharks from the Monterey Bay Aquarium. *Global Perspectives on the Biology and Life History of the Great White Shark (Carcharodon carcharias)*: 419–446.
- Sippel, T.** & Kohin, S. (2012). Catches of blue and shortfin mako sharks from U.S. West Coast recreational fisheries 1980-2010. *International Scientific Committee for Tuna and Tuna-like Species in the North Pacific ISC/12/SharkWG-1/04*: 1–9.
- Teo, S., **Sippel, T.**, ... (2012). Preliminary time series for north Pacific blue and shortfin mako sharks from the U.S. West Coast drift gillnet fishery. *NOAA*: 32.
- Teo, S., **Sippel, T.**, ... (2012). Preliminary time series for the north Pacific blue and shortfin mako sharks from the US West Coast drift gillnet fishery. *ISC/12/SHARKWG-1/03*.
- Evans, K., Kolody, D., ... **Sippel, T.** (2012). Spatial dynamics of swordfish in the south Pacific Ocean inferred from tagging experiments. | WCPFC Meetings. *Western and Central Pacific Fisheries Commission*: 36.
- Sippel, T.**, Wraith, J., ... (2011). A summary of blue shark (*Prionace glauca*) and shortfin mako shark (*Isurus oxyrinchus*) tagging data available from the North and Southwest Pacific Ocean. *NOAA*: 7.
- Sippel, T.**, Holdsworth, J., ... (2011). Investigating Behaviour and Population Dynamics of Striped Marlin (*Kajikia audax*) from the Southwest Pacific Ocean with Satellite Tags. *PLoS ONE* 6: e21087.
- Sippel, T.**, Holdsworth, J., ... (2010). Behaviour and distribution of striped marlin (*Kajikia audax*) in the southwest Pacific Ocean. *PLOS One*.
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- Holdsworth, J., **Sippel, T.** & Block, B. (2009). Near real time satellite tracking of striped marlin (*Kajikia audax*) movements in the Pacific Ocean. *Marine Biology* 156: 505–514.
- Sippel, T.**, Holdsworth, J. & Saul, P. (2009). Striped marlin satellite tagging in New Zealand: Summarizing the Foundation's research program. *New Zealand Marine Research Foundation*: 17.
- Holdsworth, J. & **Sippel, T.** (2008). A summary of New Zealand southern bluefin tuna electronic tagging results. *New Zealand Ministry of Fisheries*: 25–25.
- LePort, A., **Sippel, T.** & Montgomery, J. (2008). Observations of mesoscale movements in the short-tailed stingray (*Dasyatis brevicaudata*) from New Zealand using a novel PSAT attachment method. *Journal of Experimental Marine Biology and Ecology* 359: 110–117.
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